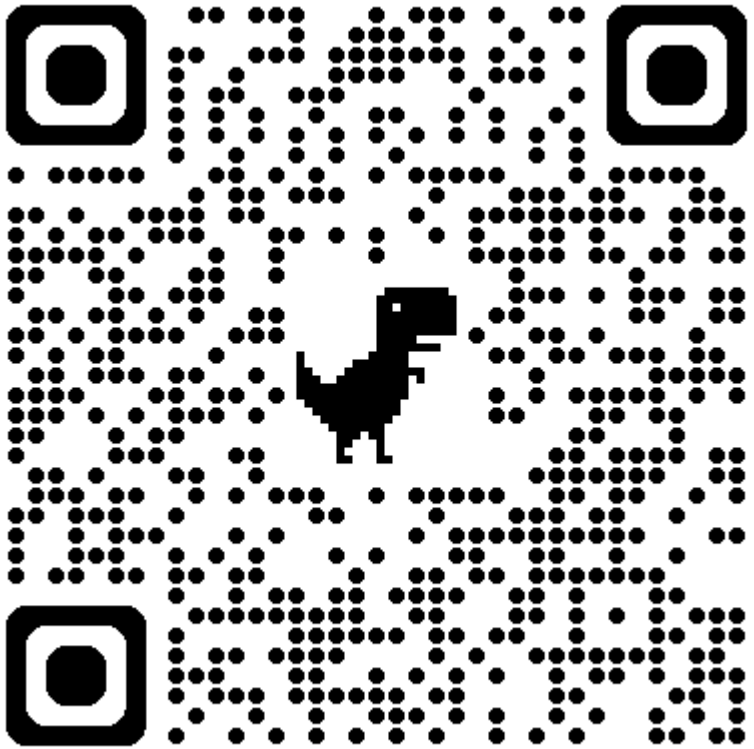
Introduction to Scientific Computing

ENGR1050 – Lecture 21

Fall 2025

Nat Trask

University of Pennsylvania



Final projects

Numerical optimization

Penn

University of Pennsylvania



Final projects

Get orders in now

Numerical optimization

Precursor to machine learning on Wednesday

Final projects

Groups of 1-4, $~25/person budget

Reports turned in by end of finals, no presentations to maximize flexibility

Final projects are *mandatory*, no using lowest exam drop to skip final project

Project

Invent something!

Order 1-2 new components

Use Gemini to teach yourself how to wire and use components

Examples of sensors



BME280
Pressure/temp/
humidity sensor
~$16



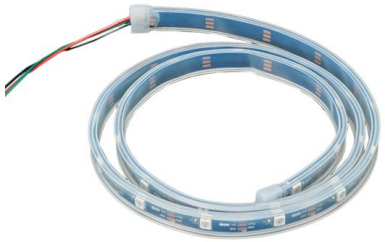
WEA012864DBP
P3N00003
LCD display
~$8



6049
Digital microphone
~$5



HC-SR04
Ultrasonic rangefinder
~$7



1528-4801-ND
Targetable LED strip
~$13



3885
Speaker
~$8



SEN-101
Color detector
~$8



SG90
Microservo motor
~$4



Arcade button
Same as our buttons
~$6

Expected difficulty: Gemini/GPT are your friends

what about a tiny lcd screen that can be used to push text

A simple option is a **0.96-inch I2C OLED (SSD1306)** or a **1.3-inch I2C LCD (ST7735/ST7789 variants)**. Both are safe for beginners and require only four wires.

1. SSD1306 128×64 OLED (recommended)

Type
Monochrome (white-on-black), very sharp, small power draw.

Wiring
- VCC (3.3 V)
- GND
- SCL
- SDA

could i use that plus 4 buttons to make pong

Thought for 13s

Yes. A Pico, a tiny LCD/OLED, and 4 buttons are enough for a simple two-player Pong.

Hardware

- **Display**: pick a 3.3 V-friendly module.
  - Easiest: **SSD1306 128×64 I²C OLED** (cheap, stable FPS for simple graphics).
  - Also fine: SPI TFTs like **ST7735 160×128** or **ILI9341 320×240**; wire via SPI (CS, DC, RST, SCK, MOSI, optional MISO).
- **Buttons (4 total)**: P1 up/down and P2 up/down. Wire each button between a **GPIO pin** and **GND**. Use Pico's **internal pull-ups** in code. No resistors needed.

Wiring example (SSD1306 I²C):

- SSD1306 VCC → 3V3, GND → GND, SDA → GP0, SCL → GP1.
- Buttons → GP2, GP3, GP4, GP5 to GND

MicroPython example (SSD1306 Pong, 2 players, 4 buttons)

Copy this entire block to `main.py` on the Pico. If your display address is 0x3C and wired to GP0/GP1, it runs as is.

python — Copy code

```
# MicroPython Pong for Raspberry Pi Pico + SSD1306 128x64 I2C + 4 buttons
# Wiring: SSD1306 VCC->3V3, GND->GND, SDA->GP0, SCL->GP1
# Buttons: each between pin and GND: GP2=P1_UP, GP3=P1_DOWN, GP4=P2_UP, GP5=P2_DO

import machine, time, math

# ----- Display setup -----
I2C_SDA = 0
I2C_SCL = 1
i2c = machine.I2C(0, sda=machine.Pin(I[illegible]SDA), scl=machine.Pin(I2C_SCL), freq=400
```





randomnerdtutorials.com/raspberry-pi-pico-bme280-micropython/?utm_source=chatgpt.com

Grants | TRASK, Nathaniel -... | natrask/PIMI-Lab: L... | Google AI studio | PIMILab - Home | OneStop Portal | FM Resource Sched...

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earn Raspberry Pi Pico

aspberry Pi Pico Pinout

licroPython

etting Started

rogramming with VS ode

nputs and Outputs

nalog Inputs

WM

HT11/DHT22

S18B20

ME280

uilt-in Temp. Sensor

SD1306 OLED

CWL-0516

IR Motion

2C LCD Display

Raspberry Pi Pico: BME280 Get Temperature, Humidity, and Pressure (MicroPython)

Learn how to use the BME280 sensor module with the Raspberry Pi Pico board with MicroPython to get temperature, humidity, and pressure readings. This is a short tutorial to get you started quickly interfacing the BME280 sensor with the Raspberry Pi Pico board.

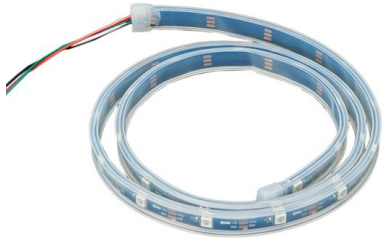


Some ideas

Audio visualizer



6049
Digital microphone
~$5



1528-4801-ND
Targetable LED strip
~$13

Theramin



HC-SR04
Ultrasonic rangefinder
~$7



3885
Speaker
~$8

Weather station



BME280
Pressure/temp/
humidity sensor
~$16



WEA012864DBPP3N00003
LCD display
~$8

Some ideas



https://randomnerdtutorials.com/projects-raspberry-pi-pico

https://how2electronics.com/raspberry-pi/raspberry-pi-pico-w-projects

https://www.dusuniot.com/blog/top-raspberry-pi-blogs

Grading Criteria

Independently make something

- Make a video showing that you figured out how to make your components go (just like in lab)
- Must have python (even if a simple while loop to make circuit go)
- No penalty for ambition
- Build up your portfolio of design projects

Orders in Wednesday

Report due by end of finals

Other options (no ordering required)

- Make the 1D helicopter buttons so it goes up and down
- Generate/find a dataset and use machine learning on it
- Any other programming project is fair game, just run by Prof. Trask in OH to help dial in difficulty

Numerical Optimization

Newton's method to solve nonlinear equations

Goal: Find x^* such that $f(x^*) = 0$

The Algorithm:

1. Start with an initial guess x_0
2. Iterate using:

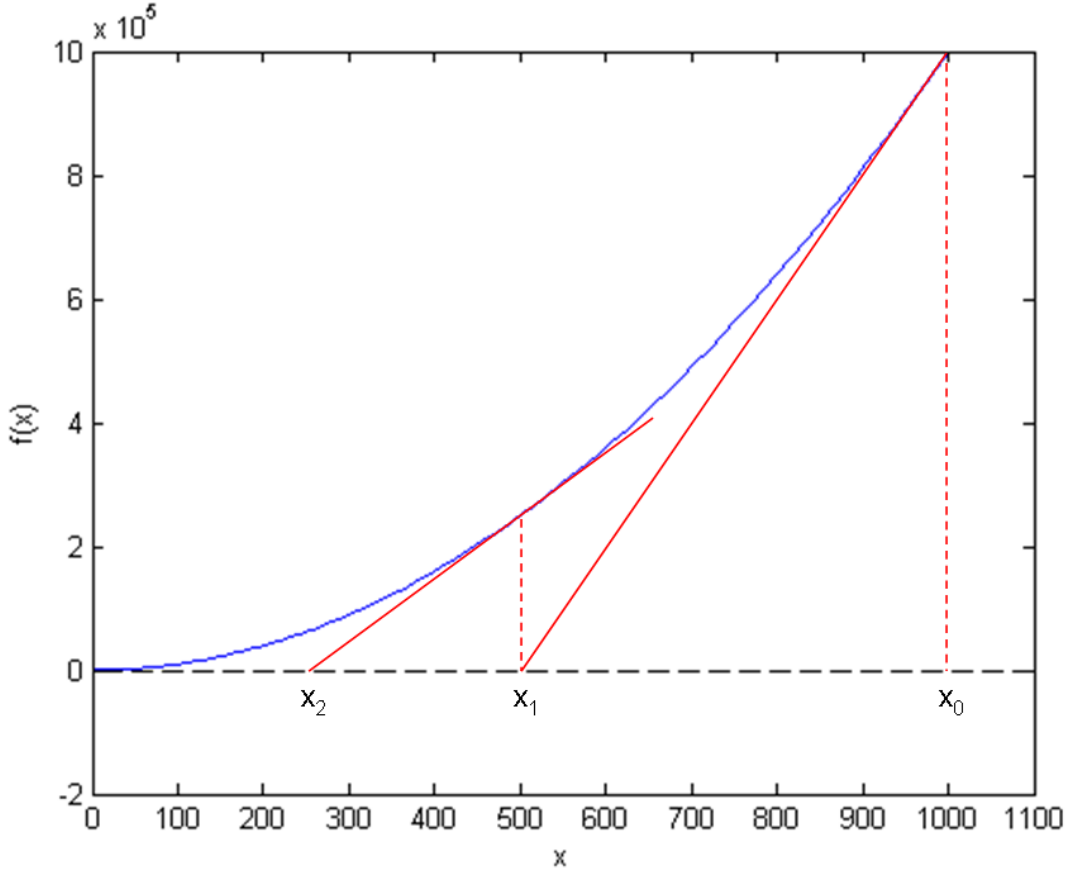
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

3. Stop when $|x_{n+1} - x_n| <$ tolerance

Geometric Intuition:

- Draw tangent line at current point $(x_n, f(x_n))$
- Find where tangent crosses x-axis
- Use that as next guess x_{n+1}

Note: Requires differentiable $f(x)$ and good initial guess



Newton is not guaranteed to converge!

Common Failure Modes

1. **Derivative is zero:** $f'(x_n) = 0$ makes update undefined
2. **Poor initial guess:** May converge to wrong root or diverge
3. **Oscillation:** Iterates bounce back and forth
4. **Flat regions:** Small derivative causes huge jumps

Example 1: $f(x) = \sqrt[3]{x}$

- $f'(0) = \infty$ (undefined)
- Near $x = 0$, Newton diverges
- Iterates: $x_1 = -2x_0$

Example 2: $f(x) = x^3 - 2x + 2$

- Local max/min can trap iterates
- Starting at $x_0 = 0$:
- $x_1 = 1, x_2 = 0, x_3 = 1, \ldots$
- Oscillates forever!

Solution: Use safeguards (max iterations, step limits) or hybrid methods

Supervised learning

Goal: Given data points (x_i, y_i), find parameters θ that best fit a model

The Problem

- Model: $\hat{y} = f(x; \theta)$ (prediction)
- Data: $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ (observations)
- Loss function: $L(\theta) = \sum_{i=1}^{n} (y_i - f(x_i; \theta))^2$ (error)

Optimization Approach:

1. Minimize loss: find θ^* where $\frac{\partial L}{\partial \theta} = 0$
2. This is root finding! Apply Newton's method
3. Or use optimization form: $\theta_{n+1} = \theta_n - \frac{\partial L/\partial \theta}{\partial^2 L/\partial \theta^2}$

Connection to Engineering:

- Fit helicopter control model to flight data
- Calibrate COVID model (β, γ) to case counts
- Train neural networks (gradient descent is simplified Newton)

Applying Newton to optimize a function

Root Finding:

- Find $f(x) = 0$
- Update:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Optimization:

- Find $f'(x) = 0$
- Update:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

Example: Minimize $f(x) = x^2 - 4x + 5$

- $f'(x) = 2x - 4$
- $f''(x) = 2$
- Update:

$$x_{n+1} = x_n - \frac{2x_n - 4}{2}$$

- Starting from $x_0 = 0$:
  - $x_1 = 0 - \frac{-4}{2} = 2$
  - Converges in 1 step!
- Minimum at $x^* = 2$

Note: Newton finds any critical point (min, max, or saddle)

Today's exercise: implement Newton, use the isoperimetric problem to check

- **Given:** Fixed perimeter P
- **Find:** Shape with maximum area

For Rectangles:

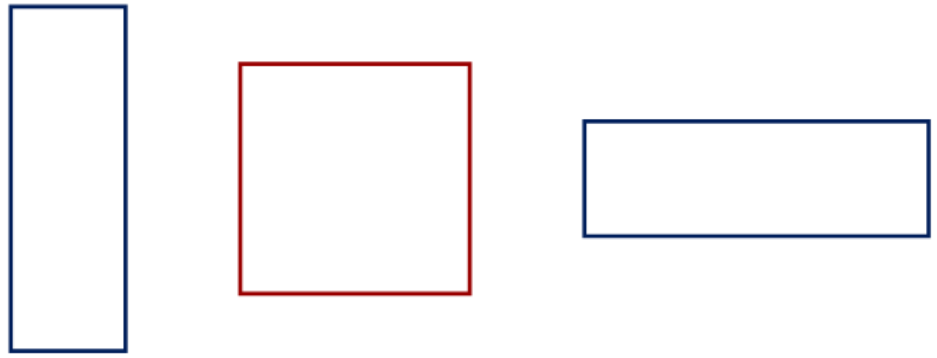
- Perimeter: $P = 2w + 2h$
- Area: $A = wh$
- Constraint: $h = \frac{P}{2} - w$
- So: $A(w) = w(\frac{P}{2} - w)$

Optimization:

$$A'(w) = \frac{P}{2} - 2w = 0$$

$$\Rightarrow w^* = \frac{P}{4} = h^*$$

All have perimeter $P = 8$



$A = 3$ $A = 4$ **(max)** $A = 3$

Newton's Method Test:

- $f(w) = A'(w) = \frac{P}{2} - 2w$
- $f'(w) = -2$
- Update: $w_{n+1} = w_n - \frac{f(w_n)}{f'(w_n)}$

Next homework out, due Monday

